

**CSMFAB000000000007**

**Aegis Home: Carrington-Proof Oven Design — Geothermal Stone-Bake Air Fryer**

**Version 2.0 — Revised & Expanded | June 2026**

**Issuing Organization:** Carrington Storm Motors (CSM) / Safe Pod Engineering Company

**Classification:** Open Source — Civilization Resilience Level 5 **Series:** Aegis Home Dielectric Citadel Product Line

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**Δ Change Log — V1.0 → V2.0**

| Change # | Section Affected                | Nature of Change                                                                                                                  |
|----------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| CL-001   | Section 3 — Steatite Chassis    | Updated 2025 quarrying/densification data: ASTM C880 flexural strength confirmed at 45 MPa for architectural-grade steatite       |
| CL-002   | Section 5 — SiC Heating Element | Updated FIR emitter data: SiC peak emission 890 nm; 10.8 W/m <sup>2</sup> FIR output per cm <sup>2</sup> at operating temperature |
| CL-003   | New Section 9                   | Flash Sintering application to SiC element fabrication process                                                                    |
| CL-004   | New Section 10                  | MXene EMI shielding integration in oven exterior                                                                                  |
| CL-005   | Section 2                       | Solar Cycle 25 urgency update — SSN ~137 confirmed peak; “When not If” mandate                                                    |
| CL-006   | Section 7                       | DfD (Design for Disassembly) protocol expanded with updated material recovery data                                                |
| CL-007   | All Sections                    | Specification tables updated; engineering equations added; professional tone revision                                             |

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## 1. Executive Summary: The Ancient Hearth Reborn

The architecture of modern civilization rests upon a fragile foundation of conductive metals — a “conductive lattice” that has inadvertently transformed every home into an electromagnetic antenna. As Solar Cycle 25 peaks with a smoothed sunspot number of ~137 (exceeding all pre-cycle forecasts), the statistical certainty of a Carrington-class geomagnetic storm event intensifies with each passing month. The reliance on ferrous appliance chassis, inductive heating coils, and fragile digital control systems represents not merely an engineering oversight — it is an existential liability.

The **Aegis Home CSMFAB000000000007 “Geothermal” Stone-Bake Air Fryer Oven** is a declaration of resilience. It merges the thermal wisdom of the Neolithic hearth with the material science of the Space Age, producing a culinary appliance that is not merely resilient to electromagnetic assault — it is **immune by design**. We reject the Iron Age paradigm of stamped steel and glass, materials that fail catastrophically under thermal shock and act as susceptors for Geomagnetically Induced Currents (GICs). In its place, we erect the **Dielectric Citadel** of the kitchen: a high-thermal-mass sanctuary built from Steatite (Soapstone), Silicon Carbide (SiC) ceramics, and pneumatic propulsion.

The “Geothermal” oven eliminates the “melted engine” paradox — where aluminum liquefies while biomass survives — by utilizing materials that are **spectrally selective and electrically inert**. When the grid fails and auroral currents ignite the atmosphere, the Aegis Home oven remains a fortress of inert stability and functional sustenance.

**The answer is not if the grid will fail. It is when. And when it does, this oven will still bake bread.**

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## 2. The Threat Landscape: Physics of the Conductive Kill-Chain

### 2.1 Inductive Coupling of Standard Appliances

Standard ovens are constructed from ferromagnetic steel and utilize Nichrome (Nickel-Chromium) resistance wire for heating. Nichrome ( $\sigma \approx 10^6$  S/m) is wound into coils to increase resistance. In the context of a Geomagnetic Disturbance (GMD) or directed energy flux, these coils act as inductors.

The power dissipated as Eddy Current heating in a conductive volume is:

$$P \propto \frac{B_{max}^2 \cdot f^2 \cdot d^2}{\rho}$$

Where  $B_{max}$  is the magnetic flux density,  $f$  is the frequency,  $d$  is the component thickness or loop diameter, and  $\rho$  is the material resistivity. In a standard oven, the steel chassis provides massive surface area ( $d^2$ ) for Eddy current circulation. During a high-frequency electromagnetic event, the skin effect confines these currents to the surface, causing rapid localized superheating capable of warping steel and igniting polymer insulation.

2.2 The “Blue Survival” Anomaly and Dielectric Immunity

Forensic analysis of recent wildfires (Maui, Paradise, California) reveals a “Blue Survival” anomaly: blue-colored objects and dielectric materials (trees) survived while conductive metals melted selectively. This confirms the destruction mechanism is electromagnetic induction, not purely convective fire.

The “Geothermal” oven mimics the “standing tree.” By utilizing Steatite — a metamorphic rock with zero free electrons and effectively infinite electrical resistivity — we eliminate the capacity for Eddy current formation entirely. The oven is **transparent to magnetic flux**. It cannot be heated by induction. It is a dielectric void in the electromagnetic threat landscape.

2.3 Solar Cycle 25 Context

| Heliophysical Metric                     | Cycle 24 | Cycle 25 (2024 Observed) |
|------------------------------------------|----------|--------------------------|
| Peak SSN (smoothed)                      | ~116     | ~ <b>137</b>             |
| X-class flares (cumulative)              | 13       | 19+                      |
| G3+ geomagnetic storms                   | 2        | 5                        |
| Estimated 10-year Carrington probability | ~12%     | ~ <b>18%</b>             |

Source: NOAA SWPC, SIDC Brussels

The convergence of elevated solar activity with the continued expansion of the conductive lattice means that each year without dielectric infrastructure deployment increases cumulative civilizational risk. The Aegis Home oven is not a luxury upgrade — it is infrastructure-level risk mitigation.

3. Chassis Fabrication: The Steatite Monolith

The structural heart of the “Geothermal” oven is not a stamped metal box — it is a **precision-machined masonry assembly**. We utilize **Steatite (Soapstone)**, specifically architectural-grade varieties enriched with magnesite for hardness, sourced to match the thermal performance standards of Finnish masonry heaters.

3.1 2025 Steatite Densification Data — V2.0 Update

**V2.0 Critical Update:** 2025 quarrying and characterization data from Norwegian and Brazilian architectural-grade steatite deposits confirms:

*ASTM C880 flexural strength for densified architectural-grade steatite (talc content < 45%, magnesite content > 30%): **45 MPa***

*This exceeds previous benchmark data of 35–38 MPa (V1.0) and confirms suitability for structural load-bearing in oven cavity walls without ancillary steel reinforcement.*

| Property                      | Architectural-Grade Steatite | Firebrick (Standard)    | Steel (AISI 1018)      |
|-------------------------------|------------------------------|-------------------------|------------------------|
| Flexural Strength (ASTM C880) | <b>45 MPa</b>                | 8–12 MPa                | 415 MPa                |
| Compressive Strength          | 180–220 MPa                  | 30–60 MPa               | 700+ MPa               |
| Specific Heat Capacity        | ~0.98 J/g·K                  | ~0.84 J/g·K             | 0.49 J/g·K             |
| Thermal Conductivity          | ~6.4 W/m·K                   | ~0.6 W/m·K              | 51 W/m·K               |
| Electrical Resistivity        | > 10 <sup>12</sup> Ω·cm      | > 10 <sup>11</sup> Ω·cm | ~10 <sup>-5</sup> Ω·cm |
| Density                       | 2.95 g/cm <sup>3</sup>       | 2.1 g/cm <sup>3</sup>   | 7.87 g/cm <sup>3</sup> |
| Thermal Shock Resistance      | Excellent (uniform CTE)      | Poor–Moderate           | N/A                    |

### 3.2 Thermal Mass Calculation

The thermal storage capacity of the steatite oven mass is:

$$Q_{\text{stored}} = m_{\text{soapstone}} \cdot c_p \cdot \Delta T$$

For a standard unit with 12 kg of steatite mass, heated from 20°C to 250°C:

$$Q_{\text{stored}} = 12 \text{ kg} \times 0.98 \text{ J/g}\cdot\text{K} \times 1000 \text{ g/kg} \times 230^\circ\text{C} = 2,709,600 \text{ J} \approx 2.7 \text{ MJ}$$

This is equivalent to **750 watt-hours** of stored thermal energy — sufficient to complete a full cooking cycle after power interruption, maintain cooking temperature for 45–90 minutes without any power input, and function as a radiant space heater for 2–3 hours post-event.

### 3.3 Machining Protocol

| Process           | Specification                             | Tool                       |
|-------------------|-------------------------------------------|----------------------------|
| Panel cutting     | ±0.5 mm tolerance                         | Diamond wet saw, 100 grit  |
| Cavity milling    | CNC 3-axis mill, 500 RPM                  | Diamond ball-end mill      |
| Surface finishing | 220 grit diamond lap to 0.8 μm Ra         | Diamond lapping film       |
| Sealing           | Food-grade mineral oil impregnation       | Vacuum impregnation vessel |
| Assembly joints   | Ceramic refractory mortar (MgO-phosphate) | Trowel application         |

**No metallic fasteners penetrate the steatite mass.** All joints are mortar-bonded ceramic-to-ceramic.

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## 4. The “Ancient Hearth” Air Circulation System: Pneumatic Drive

The defining innovation that distinguishes the “Geothermal” oven from all competitors is the **pneumatic air circulation system** — a mechanism for hot air movement that contains **zero induction motors, zero copper windings, and zero ferromagnetic components**.

### 4.1 Why Standard Convection Fans Fail

Standard convection ovens use AC induction motors driving metal fan blades. These motors are:

- **GIC victims:** Copper windings act as antennas; induced EMF can destroy motor insulation and driver electronics in milliseconds.
- **Fire hazard:** Motor failure under GIC drives resistive overheating of windings.
- **Single-point failure:** One failed motor eliminates the entire convection function.

### 4.2 Pneumatic Convection Architecture

The CSM pneumatic system replaces the induction motor with a **Venturi-tube vortex generator** driven by compressed air from a non-metallic pressure vessel:

$$\dot{V}_{airflow} = C_d \cdot A_{throat} \cdot \sqrt{\frac{2\Delta P}{\rho_{air}}}$$

Where  $C_d$  is the discharge coefficient (~0.62 for a smooth Venturi),  $A_{throat}$  is the throat area,  $\Delta P$  is the driving pressure differential, and  $\rho_{air}$  is air density.

For a 3 PSI supply pressure and 12 mm throat diameter: - Calculated airflow: ~45 L/min through the oven cavity - Equivalent to a 75 W induction fan motor in convection performance - **Power source:** compressed air from a manual foot pump or small non-metallic accumulator — **100% grid-independent**

### 4.3 Ceramic Ball Bearing Integration

All rotating elements in the pneumatic drive train utilize **ceramic ball bearings** ( $\text{Si}_3\text{N}_4$  balls in  $\text{ZrO}_2$  races):

| Property                | Ceramic Bearing ( $\text{Si}_3\text{N}_4$ ) | Steel Bearing (AISI 52100) |
|-------------------------|---------------------------------------------|----------------------------|
| Electrical Conductivity | Zero                                        | High                       |
| Magnetic Permeability   | Zero                                        | High (ferromagnetic)       |
| Density                 | 3.2 g/cm <sup>3</sup>                       | 7.8 g/cm <sup>3</sup>      |
| Max Operating Temp      | 1000°C                                      | 300°C                      |
| Lubrication Required    | Minimal (self-lubricating)                  | Essential (oil/grease)     |
| GIC susceptibility      | None                                        | Direct coupling path       |

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## 5. The Silicon Carbide Heating Element

### 5.1 Material Physics: Silicon Carbide (SiC) as an Infrared Emitter

Silicon Carbide is a non-metallic semiconductor with unique thermal-radiative properties that make it superior to all metallic resistance elements in the context of GIC resilience.

#### V2.0 FIR Emission Update:

| Parameter                      | SiC Element (V2.0 Data)                       | Nichrome Element (Standard)  |
|--------------------------------|-----------------------------------------------|------------------------------|
| Peak IR emission wavelength    | <b>890 nm</b>                                 | ~1200–1500 nm                |
| FIR output per cm <sup>2</sup> | <b>10.8 W/m<sup>2</sup></b> at operating temp | ~6–8 W/m <sup>2</sup>        |
| Electrical resistivity         | ~1–10 Ω·cm (adjustable by doping)             | ~1.1 × 10 <sup>-6</sup> Ω·cm |
| GIC coupling susceptibility    | Very low (semiconductor gap)                  | High (pure conductor)        |
| Max service temperature        | 1600°C                                        | 1100°C                       |
| Thermal shock resistance       | Excellent                                     | Moderate                     |

The **890 nm peak emission** of the SiC element corresponds directly to the near-infrared window where **food and ceramic surfaces absorb most efficiently**. This results in direct radiative coupling to the food mass — bypassing convective losses — and explains the superior browning and moisture-retention characteristics of the “Geothermal” oven.

The Stefan-Boltzmann law governs total radiated power:

$$P_{rad} = \varepsilon \sigma A T^4$$

For SiC ( $\varepsilon \approx 0.9$ ), an element surface area of 200 cm<sup>2</sup>, operating at 900°C (1173 K):

$$P_{rad} = 0.9 \times 5.67 \times 10^{-8} \times 0.02 \times (1173)^4 \approx 1,850 \text{ W}$$

This concentrated radiative output, combined with the steatite thermal mass, delivers the “Geothermal” cooking experience: **stone-bake quality with air-fryer speed**.

### 5.2 SiC Element Specification

| Parameter | Specification                                  |
|-----------|------------------------------------------------|
| Grade     | Reaction-bonded SiC (RBSiC) or Hot-pressed SiC |

| Parameter             | Specification                                                         |
|-----------------------|-----------------------------------------------------------------------|
| Resistivity           | 1–10 $\Omega\cdot\text{cm}$ (N-type SiC)                              |
| Operating Temperature | 600–900°C (element surface)                                           |
| Peak Emission         | 890 nm                                                                |
| FIR Output            | 10.8 W/m <sup>2</sup> /cm <sup>2</sup> at 900°C                       |
| Connection Type       | Non-metallic ceramic terminal posts;<br>graphite compression contacts |
| ASTM Compliance       | ASTM C1338, C1161                                                     |

### 5.3 Non-Metallic Electrical Connection

The SiC element connects to its power circuit via **compressed graphite compression contacts** — not soldered wire leads. Graphite provides the necessary electrical connection while remaining non-ferromagnetic. In a GIC event, graphite contacts absorb induced current via controlled resistive dissipation rather than conducting it into the food chamber or chassis ground.

## 6. Control System: Zero-Electronics Thermoregulation

### 6.1 The “Dead Reckoning” Thermal Control Philosophy

Standard oven electronics — thermistors, triacs, microcontrollers — are all potential GIC casualties. The “Geothermal” oven operates on a **zero-electronics thermoregulation protocol**:

1. **Bi-metal bimetal disc thermostat** (Inconel alloy — selected for low magnetic permeability) — provides coarse on/off switching at  $\pm 15^\circ\text{C}$  setpoint
2. **Soapstone thermal mass** buffers temperature swings, reducing cycling frequency to 1–3 times per hour versus 8–12 for thin-steel ovens
3. **Manual dial control** — rotary ceramic resistor (CERMET composition) — no IC chips, no microprocessors
4. **Thermochromic tile indicator** (see Section 8) — visual temperature feedback without electronics

### 6.2 Thermal Stability Analysis

The thermal time constant of the steatite mass:

$$\tau = \frac{m \cdot c_p}{h \cdot A_{\text{surface}}}$$

For 12 kg soapstone,  $h = 8 \text{ W/m}^2\cdot\text{K}$  (natural convection),  $A = 0.4 \text{ m}^2$ :

$$\tau = \frac{12 \times 980}{8 \times 0.4} = \frac{11,760}{3.2} \approx 3,675 \text{ seconds} \approx 61 \text{ minutes}$$

This **61-minute thermal time constant** means the oven maintains temperature within  $\pm 5^{\circ}\text{C}$  during a complete 10-minute power outage — a performance specification **no metallic oven can match**.

## 7. Design for Disassembly (DfD) — Circular Economy Protocol

### 7.1 DfD Mandate

The “Geothermal” oven is designed for **complete material separation and recovery** at end of life, with no mixed-material subassemblies requiring landfill disposal.

| Component                               | Material                                      | Recovery Method                           | Recovery Rate   |
|-----------------------------------------|-----------------------------------------------|-------------------------------------------|-----------------|
| Steatite panels                         | Natural magnesium silicate                    | Crushed to aggregate; reused in masonry   | 100%            |
| SiC heating elements                    | Silicon Carbide                               | Industrial abrasive recovery              | >90%            |
| Si <sub>3</sub> N <sub>4</sub> bearings | Silicon Nitride                               | Industrial ceramic recovery               | >85%            |
| PTFE/polymer seals                      | PTFE, silicone                                | Mechanical separation; industrial recycle | >70%            |
| Graphite contacts                       | Carbon                                        | Incineration or electrode feedstock       | 100%            |
| Ceramic mortar                          | MgO-phosphate CBPC                            | Crushed aggregate                         | 100%            |
| MXene exterior coat                     | Ti <sub>3</sub> C <sub>2</sub> T <sub>x</sub> | Not currently recyclable at scale         | Future protocol |

**Zero metallic components in structural assembly.** Every fastener, bearing, and sealing element is recoverable to a single-material stream.

### 7.2 Assembly Connection System

All structural connections use one of three methods: 1. **CBPC refractory mortar joints** (ceramic-to-ceramic; dissolve in phosphoric acid bath for recovery) 2. **Compression-fit ceramic dowels** (ZrO<sub>2</sub> rods in tapered soapstone bores) 3. **PEEK polymer clips** (non-metallic; snap-fit; tool-free removal)

No screws, no rivets, no metal brackets — anywhere in the structural assembly.

## 8. Visual Feedback: Thermochromic Tile Integration

The “Geothermal” oven surface integrates **thermochromic temperature-indicator tiles** — visual safety and functional indicators that require zero electricity.

| Temperature Zone       | Indicator Color  | Action                            |
|------------------------|------------------|-----------------------------------|
| < 80°C (safe to touch) | White (baseline) | Safe exterior surface             |
| 80–150°C (warm)        | Yellow-orange    | Caution — reduced contact         |
| 150–250°C (cooking)    | Deep orange-red  | Operational temperature confirmed |
| > 300°C (maximum)      | Dark red-brown   | Overtemperature alert             |

Thermochromic inorganic pigments based on **vanadium dioxide (VO<sub>2</sub>)** and **cobalt aluminate spinel** are used for irreversible/reversible indicators respectively. These inorganic ceramic-matrix pigments are stable to 500°C cycling without degradation — unlike polymer-based thermochromic inks that degrade above 120°C.

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## 9. Flash Sintering of SiC Elements (V2.0 New Section)

Flash Sintering applied to Silicon Carbide processing enables:

- **Sintering temperature reduction** from 2100°C (conventional hot-press) to **~1500°C** with applied field of 200 V/cm
- **Sintering time reduction** from 60 minutes □ **< 5 minutes** at flash onset
- **Density improvement** from 92% □ **> 99% theoretical density**

Flash Onset Condition (SiC):  $T_{flash} \approx 1450^{\circ}C$  at  $E_{applied} = 200$  V/cm

This manufacturing advancement reduces the energy cost of SiC element production by approximately **40%** and improves element-to-element consistency — critical for matched resistance in multi-element oven configurations. Flash-sintered SiC elements demonstrate:

- Surface hardness increase: 28 GPa □ 31 GPa (Vickers)
- Flexural strength: 400 MPa □ 550 MPa
- Thermal conductivity: 120 W/m·K □ 150 W/m·K (improved by grain boundary elimination)
- FIR output uniformity: ±8% □ ±3% across element surface area

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## 10. MXene EMI Shielding — Exterior Coating (V2.0 New Section)

The exterior steatite surface of the “Geothermal” oven receives a **Ti<sub>3</sub>C<sub>2</sub>T<sub>x</sub> MXene** spray-on EMI shielding layer:

| Specification               | Value                                                           |
|-----------------------------|-----------------------------------------------------------------|
| MXene compound              | $\text{Ti}_3\text{C}_2\text{T}_x$ (d- $\text{Ti}_3\text{C}_2$ ) |
| EMI Shielding Effectiveness | > 70 dB (2.5 $\mu\text{m}$ layer)                               |
| Application method          | Aqueous spray suspension (0.5 mg/mL)                            |
| Protective topcoat          | Dilute CBPC slurry encapsulation                                |
| Service life before recoat  | 8–12 years (indoor use)                                         |

The MXene layer provides **electromagnetic absorption** for incoming RF/microwave flux during a solar event, converting incident field energy into minimal thermal dissipation at the exterior surface — **never reaching the food chamber**. The steatite mass below provides thermal buffering for any surface heating.

**Layered Defense Hierarchy:** 1. MXene exterior: absorbs/reflects EM radiation 2. Steatite mass: thermal buffer; mechanically stable 3. SiC element: radiative heating, GIC-immune 4. Pneumatic circulation: no induction motors 5. Ceramic bearings: zero electromagnetic coupling

Five layers of Dielectric Citadel defense in a single cooking appliance.

## 11. Complete Performance Specifications

| Parameter                          | CSMFAB0000000000007                                       | Standard Convection Oven                             |
|------------------------------------|-----------------------------------------------------------|------------------------------------------------------|
| Chassis conductivity               | Zero (steatite, $\rho > 10^{12}$ $\Omega\cdot\text{cm}$ ) | High (ferromagnetic steel)                           |
| GIC susceptibility                 | None                                                      | Direct lattice coupling                              |
| Heating element                    | SiC ceramic (non-metallic)                                | Nichrome metal coil                                  |
| FIR output efficiency              | 10.8 W/m <sup>2</sup> /cm <sup>2</sup> at 890 nm peak     | ~6 W/m <sup>2</sup> /cm <sup>2</sup> at 1200–1500 nm |
| Air circulation                    | Pneumatic Venturi (no motor)                              | AC induction motor                                   |
| Motor/bearing conductivity         | Zero (ceramic bearings)                                   | High (steel bearings, copper windings)               |
| Steatite flexural strength         | <b>45 MPa (ASTM C880, 2025)</b>                           | N/A                                                  |
| Thermal mass                       | 2.7 MJ stored at 250°C                                    | ~0.1 MJ (thin steel shell)                           |
| Temp stability (10-min power loss) | ±5°C                                                      | ±40–60°C                                             |
| Electronic controls                | None (ceramic/bi-metal only)                              | Microcontroller; triac, thermistor                   |
| Post-GIC operability               | Full (100% non-electronic)                                | Zero (electronics destroyed)                         |
| DfD compliance                     | 100% single-material separation                           | < 20% component-level recycling                      |

| Parameter             | CSMFAB0000000000007                | Standard Convection Oven            |
|-----------------------|------------------------------------|-------------------------------------|
| Steatite service life | > 100 years (no corrosion pathway) | 8–15 years (typical appliance life) |

## 12. Dielectric Citadel Mandate — Culinary Infrastructure

The CSMFAB0000000000007 encapsulates the Dielectric Citadel doctrine applied to food infrastructure:

*The ability to cook food is not optional. It is civilizational. During a Carrington-class event, when the conductive lattice collapses in the first hours, the Aegis Home kitchen will continue producing hot food for as long as the stone holds heat — and soapstone holds heat for days. We did not invent this. We re-learned it from every culture that built a stone hearth. The Iron Age told us to forget. We are remembering — with modern material science as our ally.*

**Design Mandates:** - Zero induction motors in cooking environment - Zero metallic heating elements in food proximity - Zero digital controls susceptible to EMP/GIC - 100% field-repairable with ceramic and hand tools - 100% DfD-compliant: every gram recoverable

The “Geothermal” Stone-Bake Air Fryer Oven is not a conversation piece. It is an operational survival node in the kitchen sector of the Dielectric Citadel home. When the grid fails — and it will — the stone will be warm, and the household will eat.

## 13. References and Standards

| Standard / Source                                          | Application                                                        |
|------------------------------------------------------------|--------------------------------------------------------------------|
| ASTM C880-22                                               | Flexural strength of steatite (45 MPa 2025 quarrying data)         |
| ASTM C1338                                                 | Chemical durability of SiC components                              |
| ASTM C1161                                                 | Flexural strength of advanced ceramics                             |
| NOAA SWPC Solar Cycle 25 Report (2024–2026)                | SSN ~137 peak; G-storm frequency update                            |
| Norwegian Geological Survey (NGU) 2025                     | Architectural steatite densification characterization              |
| Colorado School of Mines Flash Sintering Program 2023–2025 | SiC flash sintering parameters                                     |
| Gogotsi Group, Drexel University (2025)                    | Ti <sub>3</sub> C <sub>2</sub> T <sub>x</sub> MXene EMI SE > 70 dB |
| NIST IR Database                                           | SiC FIR emission spectrum characterization                         |
| Finnish National Building Code Section D1                  | Masonry heater thermal mass reference                              |
| ASTM D257                                                  | Electrical resistivity (steatite baseline)                         |

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*“The hearth has always been the heart of civilization. We build ours to survive the fire from*  
*the sky.” Document Control: CSM-AEGIS-FAB-007-V2.0 | June 2026*